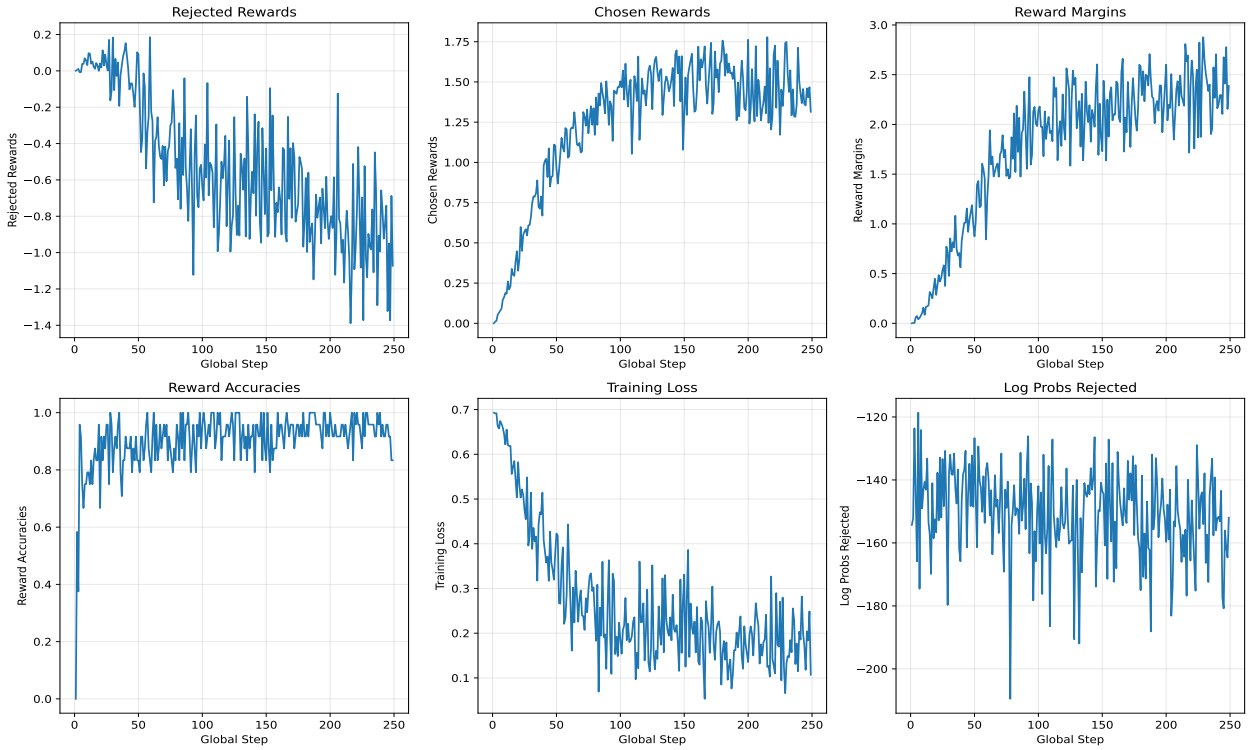
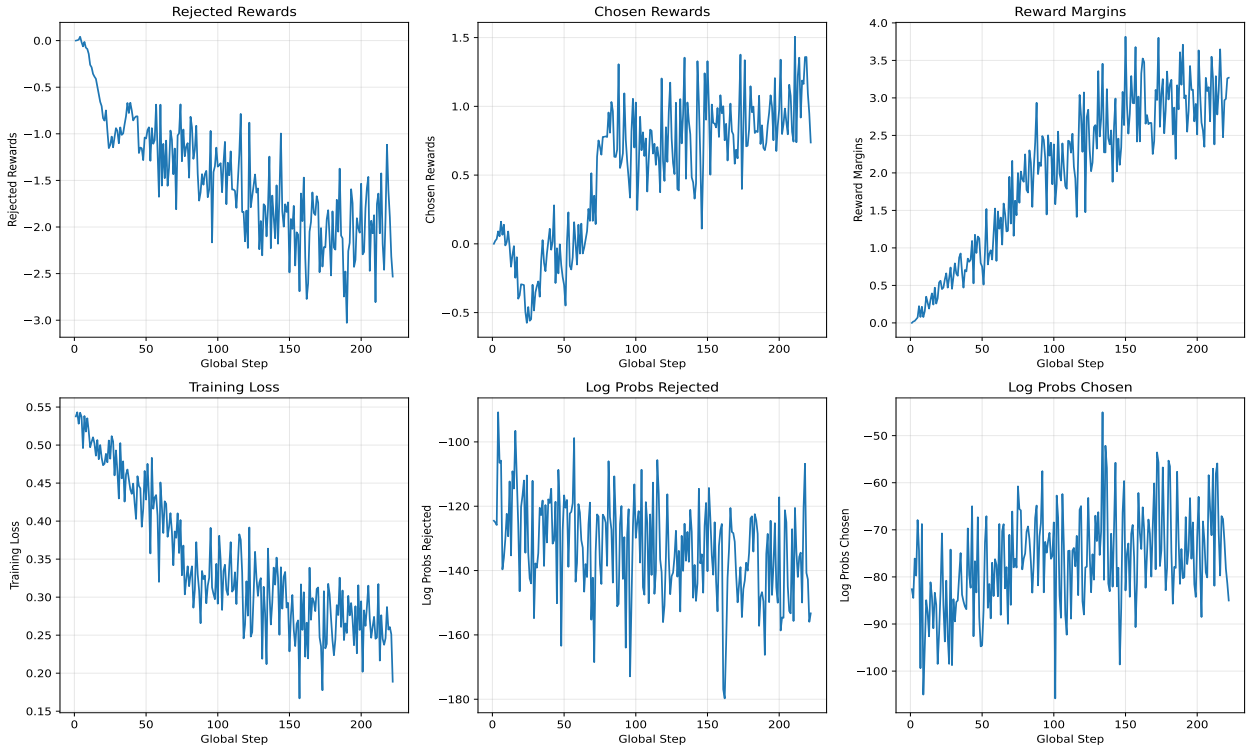
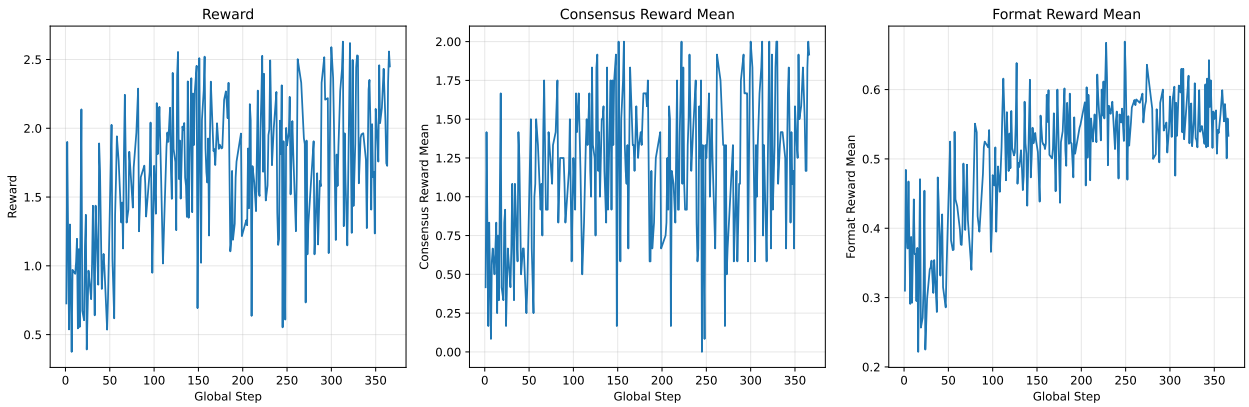


(a) MV-DPO training curves

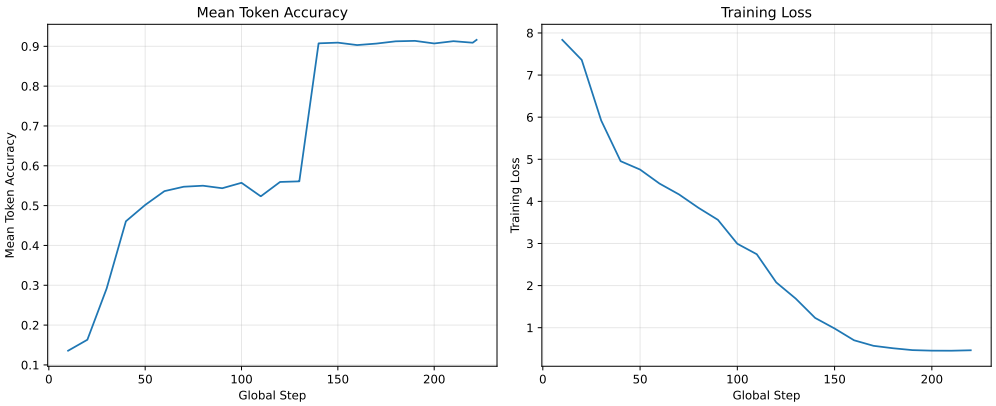


(b) MV-KTO training curves

Figure 7 Training curves for MV-DPO and MV-KTO. (a) MV-DPO: increasing reward margin between majority (chosen) and minority (rejected) responses, with declining rejected log probabilities. (b) MV-KTO: similar reward margin increase using unpaired examples, with rejected log probabilities decreasing and chosen increasing.



(a) MV-GRPO training curves



(b) MV-SFT training curves

Figure 8 Training curves for MV-GRPO and MV-SFT. (a) MV-GRPO: consensus and format rewards both increase. (b) MV-SFT: token accuracy increases while loss decreases.

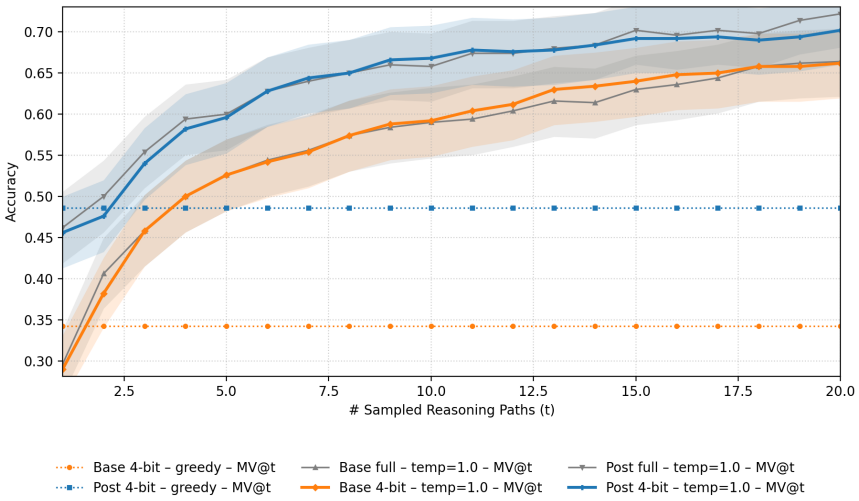


Figure 9 Llama-3B on MathQA (MV@t, t=20). Blue: post-trained 4-bit model, Orange: base 4-bit model, Grey: full-precision model.

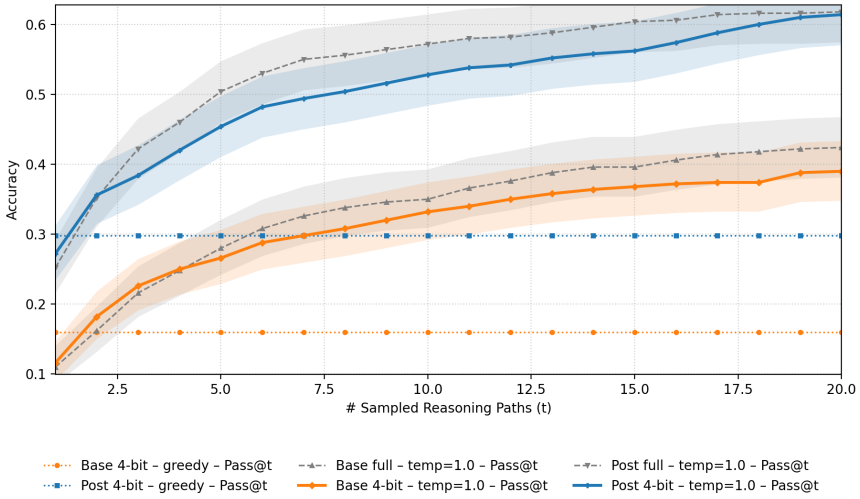


Figure 10 Qwen-2B on MATH (Pass@t, t=20). Blue: post-trained 4-bit model, Orange: base 4-bit model, Grey: full-precision model.

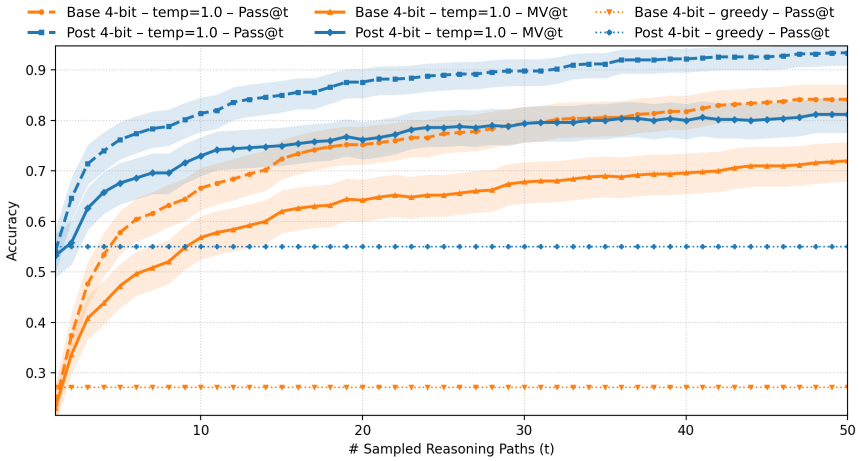


Figure 11 Qwen-2B on GSM8K (Pass@t, MV@t; t=50). Blue: post-trained 4-bit model, Orange: base 4-bit model.